DHRUV SAINI

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EXPERIENCE

A Sustainable Future

CEO | Oct 2023–Present

- Built PCM, an AI-powered mobile app that helps schools predict paper use, reaching 30,000 students nationwide in over 150 schools.
- Trained a model that finds patterns in school data to estimate paper consumption.
- Leading a team of 20 to develop a nonprofit website and improve the model; it now helps save 1.7M sheets of paper per year.
- Received $30k in grants supporting sustainability impact.

West Virginia University: SG-REAL Lab

GIS/Software Research Intern | Jul 2025–Present

- Created a 3D map and flood-detection process for power grids to identify the parts that need investment most.

Dimension

Software Engineer Intern | Jun 2025–Sept 2025

- Connected the collaboration platform to outside services through webhooks, which automatically share information between apps.
- Led support for Stripe, Vercel, GitHub, and many other hosted bots inside channels.
- Modernized the interface; the resulting refactors reduced load time by 70%.

University of Washington

Quantum Technical Writing Intern | Aug 2024–Sept 2024

- Worked with Prof. Anantram on a clearer editing process for his book Quantum Mechanics for Scientists and Engineers.
- Built Python and React tools that compare textbook drafts and help students choose the best version of any paragraph.
- Reduced editing time by 15x.

PROJECTS

CAD-bench: Benchmarking Language Models on Functional CAD Generation

- Built a test suite for AI-generated 3D parts that checks whether a part works as intended, not only whether it looks correct.
- Ran physics simulations of mechanisms such as gearboxes to test whether generated parts work in motion.
- Accepted to the ICML AIWILD and FAGEN workshops; currently under review at NeurIPS, with code available here and paper available here.

EDA Bench: An Execution-Based Benchmark for Language Model Agents on Functional PCB Construction

- Built a test suite for AI-designed circuit boards that checks electrical behavior, not only whether a design file opens.
- Combined KiCad, connection and safety checks, circuit simulation, and real-world input/output requirements to test whether a circuit functions.
- Accepted to the ICML FAGEN workshop; currently under review at NeurIPS, with code available here and paper available here.

Residual Controllers

- Studied the small internal signal that tells an open AI model whether to begin step-by-step reasoning or give a direct answer.
- On all 30 AIME 2026 problems, a one-time adjustment entered thinking mode on 30/30 and scored 16/30, compared with 1/30 when thinking was off; paper available here.

ThinkSwitch: Iterative LoRA Training and Weight Interpolation for Low-Compute Improvement on Specific-Purpose Reasoning Tasks

- Used two low-cost ways to adapt a language model—small trainable additions and blended saved model weights—to improve Qwen3-4B performance on AIME with only $3 in compute.
- Awarded 1st place at the Washington State Science and Engineering Fair; preprint available here.

Muon Browser | TypeScript, Electron

- Created a web browser that lets people place pages and notes on an open, unlimited workspace.
- Available to download and use here.

AWARDS AND HONORS

National Science Bowl

Team Captain | 2021–Present

- Founded Bellevue High School’s first science bowl team, winning 1st place at the nationwide virtual competition.
- Won a fully-funded trip to Washington, D.C. for the National Finals.

Olympiad Awards

Various | 2024–Present

- USACO Platinum Rank.
- USA AI Olympiad National Finalist, Bronze Medal.

ARC–The American Rocketry Challenge

Team Captain | 2025

- Captained a team of five to build a model rocket following specific criteria, qualified for National Finals twice.

EDUCATION

Bellevue High School

- 36 ACT, 100+ hours of community service, DECA ICDC finalist.
- President of Programming Club, coach at Tyee MS and Odle MS programming clubs.